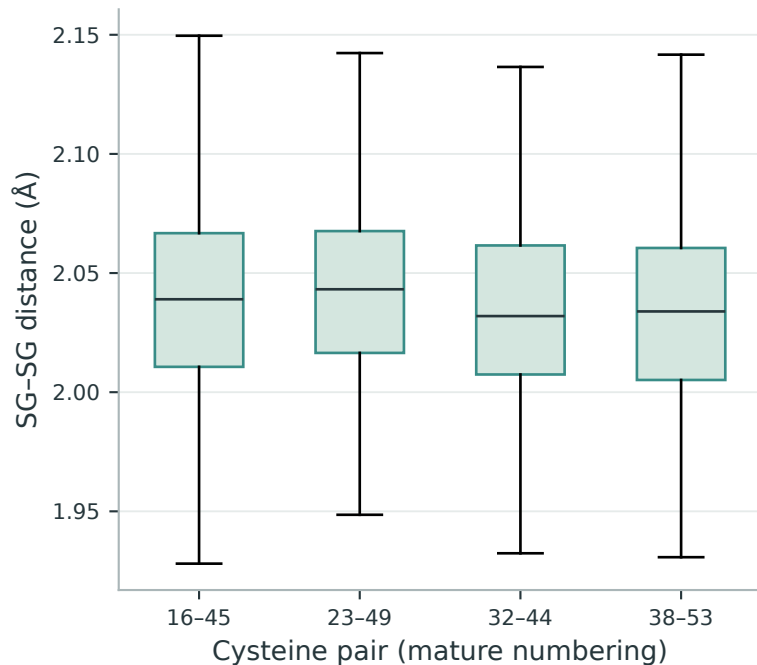


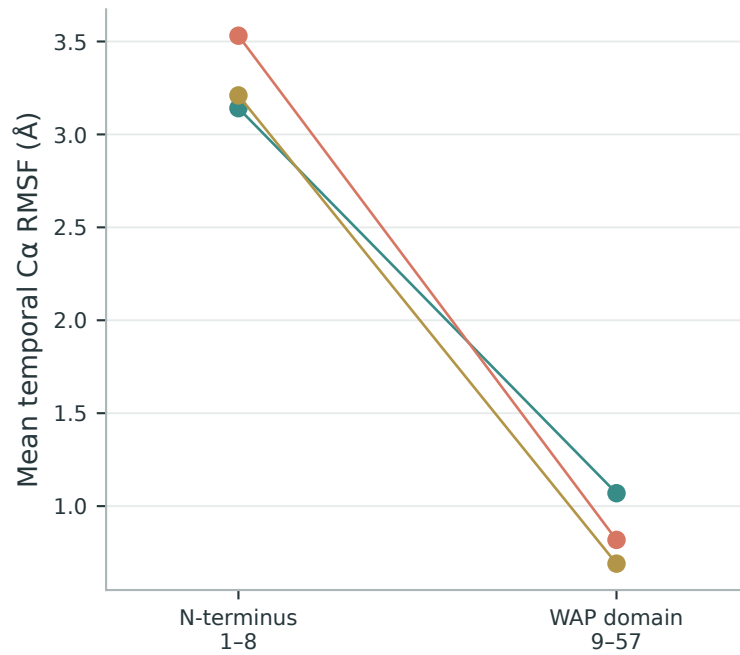
Covalent topology and short-window intramolecular contacts

Four explicit Elafin disulfide bonds | contact occupancy from completed trajectories | no receptor is present in this MD system

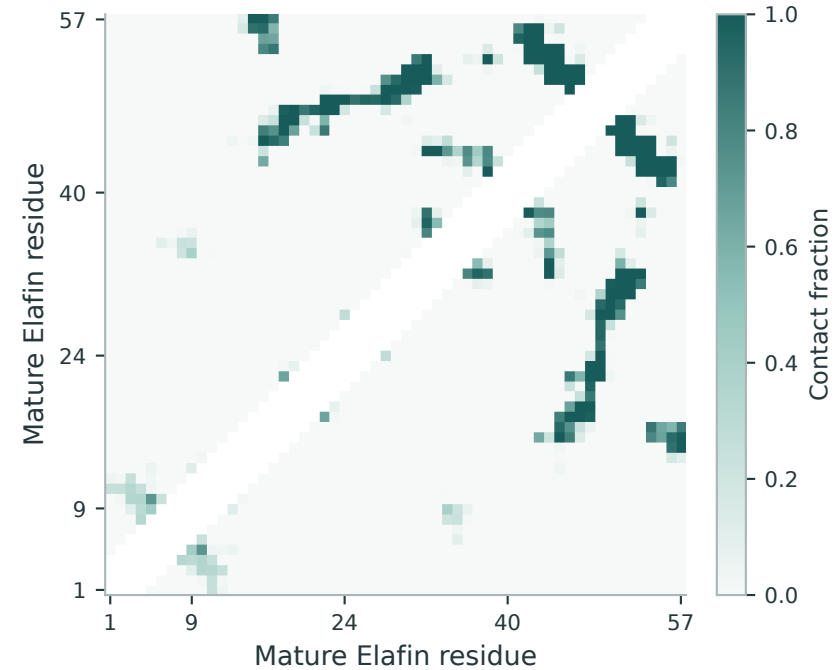
a Disulfide geometries



b Regional fluctuation contrast



c Cα contact occupancy (<8 Å)



Disulfides are bonded terms in the force field, so intact SG-SG distances are a topology check, not an independent stability assay.

Contact map: equal-weight mean of 3 completed replicas over 0.2-1.0 ns; sequence neighbours $|i-j| \leq 3$ are masked. No convergence claim.